



POWER SOLUTIONS

Operations Selection Guide

Power Solution Selection Guide for Deployment

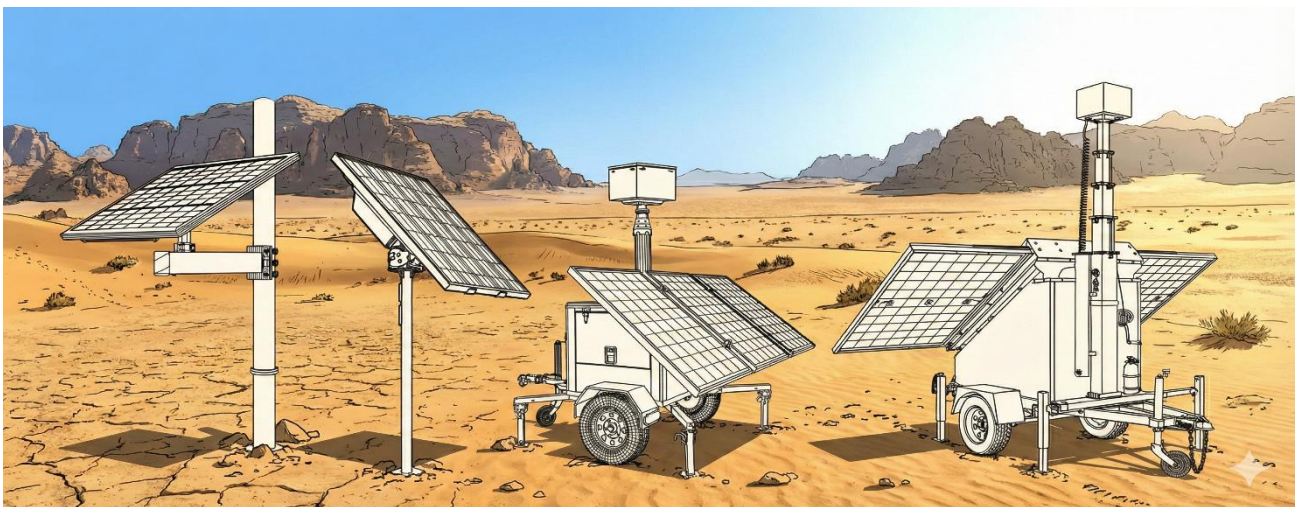


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1. Quick Selection Guide

Use this section for rapid solution selection based on your equipment load requirements.

1.1 Selection by Load

Match your total system power consumption to the appropriate Power Solution:

Total System Load*	Recommended Solution	Typical Equipment	Backup Time**
≤ 8W	WP-384	WakeCap Gateway	48 hours
9W - 15W	WP-768	Small PTZ Camera, Bullet Camera	48 hours
16W - 100W	WP-4800T	PTZ camera, bullet camera, P2P	48 hours
101W - 350W	WP-21600T	Multiple PTZ cameras, comm equipment	48 hours

* This can be increased but backup will reduce accordingly

** Backup time assumes zero solar input (worst case). Actual runtime varies with solar conditions.

⚡ IMPORTANT

Always calculate TOTAL SYSTEM POWER, not just device power. System power includes the device, PoE injectors, routers, voltage converters, and all auxiliary equipment.

1.2 Equipment Power Reference

Standard power consumption for common equipment:

Equipment Type	Device Power	System Power*	Notes
WakeCap Gateway	~5W	~8W	Includes power conversion losses
Bullet Camera (standard)	15W	~20W	Includes PoE injector, router
PTZ Camera (typical)	42W	~60W	Mid-range PTZ with PoE
PTZ Camera (with IR or other features)	62W	~80W	High-end PTZ with IR/heater
Network Router/Switch	5-15W	—	Add to system total
PoE Injector	3-8W	—	Conversion losses

* System Power = Device + PoE + Router + Conversion Losses + Misc. Always use System Power for selection.

2. Load Calculation Fundamentals

Proper load calculation is essential for selecting the correct power solution. Follow this systematic approach.

2.1 Calculation Order

Power system sizing follows this sequence:

1. **LOAD** - Calculate total continuous power consumption
2. **BATTERY** - Size battery bank for required backup duration
3. **SOLAR** - Size solar array to recharge battery daily
4. **MPPT** - Select controller to handle solar current
5. **AUXILIARY** - Add protection, cabling, mounting

2.2 Standard Design Parameters

All WakeCap Power Solutions are designed using these parameters:

Parameter	Value	Description
Backup Time	48 hours	Required autonomy without solar input
Depth of Discharge	70%	Usable battery capacity
System Efficiency	90%	Wiring and general losses
Converter Efficiency	80%	DC-DC conversion losses
Solar Derating	75%	Dust, misalignment, temperature
MPPT Efficiency	95%	Charge controller efficiency
Peak Sun Hours	6.5 hours	Saudi Arabia average

2.3 Example Calculation

Calculate the solution for a single PTZ Camera (typical) system:

- System Load: 70W (42W camera + PoE + router)
- System Voltage: 12V
- Required Battery = $(48 \times 70) / (0.70 \times 0.90 \times 0.80 \times 12)$
- Required Battery = $3360 / 6.048 = 556 \text{ Ah}$

NOTE

A 70W load requires approximately 556Ah battery capacity. The WP-4800T (400Ah) is insufficient; select WP-21600T (900Ah @ 24V = equivalent to ~450Ah usable at 12V reference, but provides 24V system) for margin. Alternatively, reduce load or accept shorter backup time.

2.5 Load Capacity Summary

Maximum continuous loads per solution (for 48-hour backup):

Solution	Battery Capacity / Power	Max Load (48h backup)	Max Load (24h backup)
WP-384	30Ah / 384Wh	~8W	~16W
WP-768	60Ah / 768Wh	~15W	~30W
WP-4800T	400Ah / 4,800Wh	~100W	~200W
WP-21600T	900Ah / 21,600Wh	~350W	~700W

3. Product Specifications Comparison

3.1 Power System Specifications

Specification	WP-384	WP-768	WP-4800T	WP-21600T
System Voltage	12V DC	12V DC	12V DC	24V DC
Battery Capacity	30Ah	60Ah	400Ah (2×200)	900Ah (6×150)
Battery Energy	384Wh	768Wh	4,800Wh	21,600Wh
Battery Type	LiFePO4	LiFePO4	GEL Lead Acid	HT Lead Acid
Solar Power	80W	100W	450W	1,305W
Solar Panels	2 × 40W	2 × 50W	3 × 150W	3 × 435W
MPPT Controller	Integrated	10A	30A	60A EPEVER
MPPT Efficiency	>95%	96.5%	95%	98%
Output Voltage	12V ±0.5V	10.5-12.6V	12V DC	24V DC
Max Output Current	3A	10A	<30A (25A typical)	<60A (55A typical)

3.2 Physical Specifications

Specification	WP-384	WP-768	WP-4800T	WP-21600T
Total Weight	~6 kg	23 kg	525 kg	1,300 kg
Form Factor	Pole/Wall Mount	Bracket Mount	Trailer	Trailer
Mast Height	N/A	N/A	6.5m (pneumatic)	9m (manual)
Transport Size (mm)	780×530×200	940×780×240	2000×1400×2300	2320×1495×2350
Deployed Size (mm)	780×530×varies	940×780×varies	2000×2250×6500	2320×1495×2350 (+ mast)
Trailer Type	-	-	Single axle	Single axle
Tow Hitch	-	-	Ball hitch	50mm ball hitch
Tire Size	-	-	165/70R13	15"

3.3 Environmental Ratings

Specification	WP-384	WP-768	WP-4800T	WP-21600T
IP Rating	IP66	IP66	IP65	IP65
Operating Temp	-20°C to +50°C	-30°C to +85°C	-20°C to +80°C	-20°C to +85°C
Charge Temp (Low)	-10°C	-30°C	[TBD]	[TBD]
Discharge Temp (Low)	-20°C	-30°C	[TBD]	[TBD]
Wind Rating (Panel)	2400Pa	60m/s (200kg/m ²)	100 km/h	100 km/h

5. Environmental Considerations

Saudi Arabia deployment environments require special attention to these factors.

5.1 Desert Environment

- High ambient temperatures (up to 50°C+)
- Solar panel derating at high temperatures (expect 10-15% reduction)
- Battery capacity reduction at extreme temperatures
- Dust accumulation on solar panels - plan monthly cleaning
- Sand/dust ingress - verify IP rating adequate for location

i NOTE

At temperatures above 45°C, expect reduced charging efficiency and shorter battery runtime. LiFePO4 batteries (WP-384, WP-768) handle heat better than Lead Acid (WP-4800T, WP-21600T). Consider shading the battery enclosure if possible.

5.2 Coastal/Humid Environment

- Salt air accelerates corrosion on exposed metal
- Higher humidity reduces solar panel efficiency slightly
- Verify all connections are sealed and protected
- More frequent maintenance inspections recommended
- Consider additional corrosion protection for trailer frames

5.3 Wind Considerations

Maximum wind ratings by product:

- WP-384/WP-768: Solar panels rated for 60 m/s (216 km/h)
- WP-4800T/WP-21600T: Mast should be lowered in high winds
- WP-21600T: System rated for 100 km/h wind speed

⚡ IMPORTANT

Always lower trailer masts before wind speeds exceed 50 km/h. Extended masts act as significant wind loads and can cause trailer instability or structural damage.

6. Installation Requirements

6.1 Solar Panel Orientation

For optimal energy harvest in Saudi Arabia:

- Panel orientation: SOUTH (0° azimuth)
- Tilt angle: 20-30° from horizontal (WP-768 adjustable 0-30°)
- Seasonal adjustment: 15° summer, 35° winter (if accessible)
- Avoid any shading, especially partial shading across cells

6.2 Trailer Setup Procedure (WP-4800T, WP-21600T)

1. Position trailer on level ground with southern sky exposure
2. Engage parking brake and chock wheels
3. Deploy outriggers and level trailer (WP-4800T)
4. Disconnect from tow vehicle
5. Deploy and orient solar panels
6. Raise mast to desired height
7. Mount equipment to mast/camera box
8. Connect equipment cables
9. Verify system operation

8. Common Deployment Scenarios

8.1 Scenario A: Remote Gateway

Situation: WakeCap Gateway for worker tracking at isolated site

- Equipment: WakeCap Gateway (~8W system)
- Solution: WP-384
- Mounting: Existing pole or temporary pole

8.2 Scenario B: Single Bullet Camera

Situation: Security camera monitoring equipment yard

- Equipment: Bullet camera + PoE + router (~20W system)
- Solution: WP-768 (15W continuous allows 48h backup at 20W)
- Mounting: Bracket on structure or dedicated pole

8.3 Scenario C: Single PTZ Camera

Situation: PTZ surveillance camera at remote checkpoint

- Equipment: PTZ camera (typical) (~70W system)
- Solution: WP-4800T (>100W capacity with margin)
- Mounting: Trailer mast (6.5m height)

8.5 Scenario E: High-Power PTZ System

Situation: High-end PTZ with heating element in harsh environment

- Equipment: PTZ Type 2 (~90W) + P2P & Switch (~25W) + Gateway (~10W) = 125W system
- Solution: WP-21600T
- Note: WP-4800T is sufficient for 125W load but with only 36 hour backup

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WakeCap Power Solutions - Operations Selection Guide v1.2
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